

Continued fractions

Continued fraction is a representation of a real number as a specific convergent sequence of rational numbers. They are useful in competitive programming because they are easy to compute and can be efficiently used to find the best possible rational approximation of the underlying real number (among all numbers whose denominator doesn't exceed a given value).

Besides that, continued fractions are closely related to Euclidean algorithm which makes them useful in a bunch of number-theoretical problems.

Continued fraction representation

Definition

Let $a_0, a_1, \dots, a_k \in \mathbb{Z}$ and $a_1, a_2, \dots, a_k \geq 1$. Then the expression

$$r = a_0 + \frac{1}{a_1 + \frac{1}{\dots + \frac{1}{a_k}}},$$

is called the **continued fraction representation** of the rational number r and is denoted shortly as $r = [a_0; a_1, a_2, \dots, a_k]$.

Example

Let $r = \frac{5}{3}$. There are two ways to represent it as a continued fraction:

$$\begin{aligned} r = [1; 1, 1, 1] &= 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1}}} \\ r = [1; 1, 2] &= 1 + \frac{1}{1 + \frac{1}{2}} \end{aligned}$$

It can be proven that any rational number can be represented as a continued fraction in exactly 2 ways:

$$r = [a_0; a_1, \dots, a_k, 1] = [a_0; a_1, \dots, a_{k+1}]$$

Moreover, the length k of such continued fraction is estimated as $k = O(\log \min(p, q))$ for $r = \frac{p}{q}$.